

Shaft Sinking Freezing Method

15/05/2020

Specialized Method of Shaft I

- *Specialized method of shaft sinking is used when following conditions are encountered.
- *Incompetent, water bearing rock (quick sand)
- *Weak unstable, soil type rocks.
- *Competent fractured rocks, with high water inflows above $0.5\text{m}^3/\text{min}$.
- *There are numerous special methods are available to suit conditions.

Shaft Thickness Frozen Wall I

*Artificially created cylinder of frozen rock around shaft creates a kind of lining, shielding inner space against water & unstable rock..

*Thickness of frozen rock cylinder for moderate depth is provided by

$$T = r * \sqrt{k/(k-2p)-1}$$

T> thickness of frozen rock cylinder

r> shaft radius in m

k> allowable stress of frozen rock Mpa

p> external pressure Mpa

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Shaft Thickness Frozen Wall II

- *Allowable stress of rock is 50% of its compressive strength in mean range of freezing temperature.
- *In well defined & favorable conditions, allowable stress can be increased to 75% of rock compressive strength.
- *Because of creep characteristic of ice, compressive strength of frozen rocks is a time dependent value.
- *For calculating thickness of cylinder of frozen rock, mean temperature of rock should be taken.

Shaft Thickness Frozen Wall III

- *Temperature within wall of frozen rock changes from minimum value just beside frizzing hole to a maximum “0” on outer wall limit.
- *Cylinder of frozen rock, when properly created, permits sinking shaft of moderate depth even in long sections using temporary lining.
- *Concrete lining length made in one pass changes from several meter to 25m.
- *It depends on strength of frozen rock mass, overall depth & effective temperature.

Shaft Thickness Frozen Wall III

*Thickness of cylinder of frozen rock at greater depth with in plastic zone is provided by Domke Formula.

Shaft Thickness Frozen Wall IV

Domke Formula

$$T/r = 0.29P/k + 2.3P/k$$

$$p < P < k$$

$$p = HDm$$

T> thickness of frozen rock cylinder

r> shaft radius in m

P> actual stress in frozen rock Mpa

k> allowable stress of frozen rock Mpa.

H> depth of frozen rock m

Dm> mean density of frozen rock

Shaft Thickness Frozen Wall V

- *For greater freezing depth, creep design concept of frozen rock thickness is applied.
- *It is usually impossible to arbitrarily change freezing temperature.
- *At pressure level, frozen rock creeps.
- *Length of unsupported shaft can stand before creeping / failure has to be determined for specific time.
- *Because of technical & economical limits, a practical thickness of frozen cylinder is assumed to be 5 to 5.5m

Shaft Thickness Frozen Wall VI

*Allowable height of excavation is to be calculated accordingly.

Shaft Thickness Frozen Wall III

Lieberman Formula

$$T_{\max} = r(D H / T_{ad}-1) S$$

$T_{\max}$ > maximum thickness of frozen rock cylinder

r > shaft radius in m

D > density t/m³

H > depth of frozen rock m

T_{ad} > adequate thickness of frozen rock cylinder to stand exposure time & creep.

S > safety factor

Shaft Thickness Frozen Wall III

Height of excavated section unsupported

$$h = T(T_{ad} / D * H)$$

h> height of excavated section unsupported

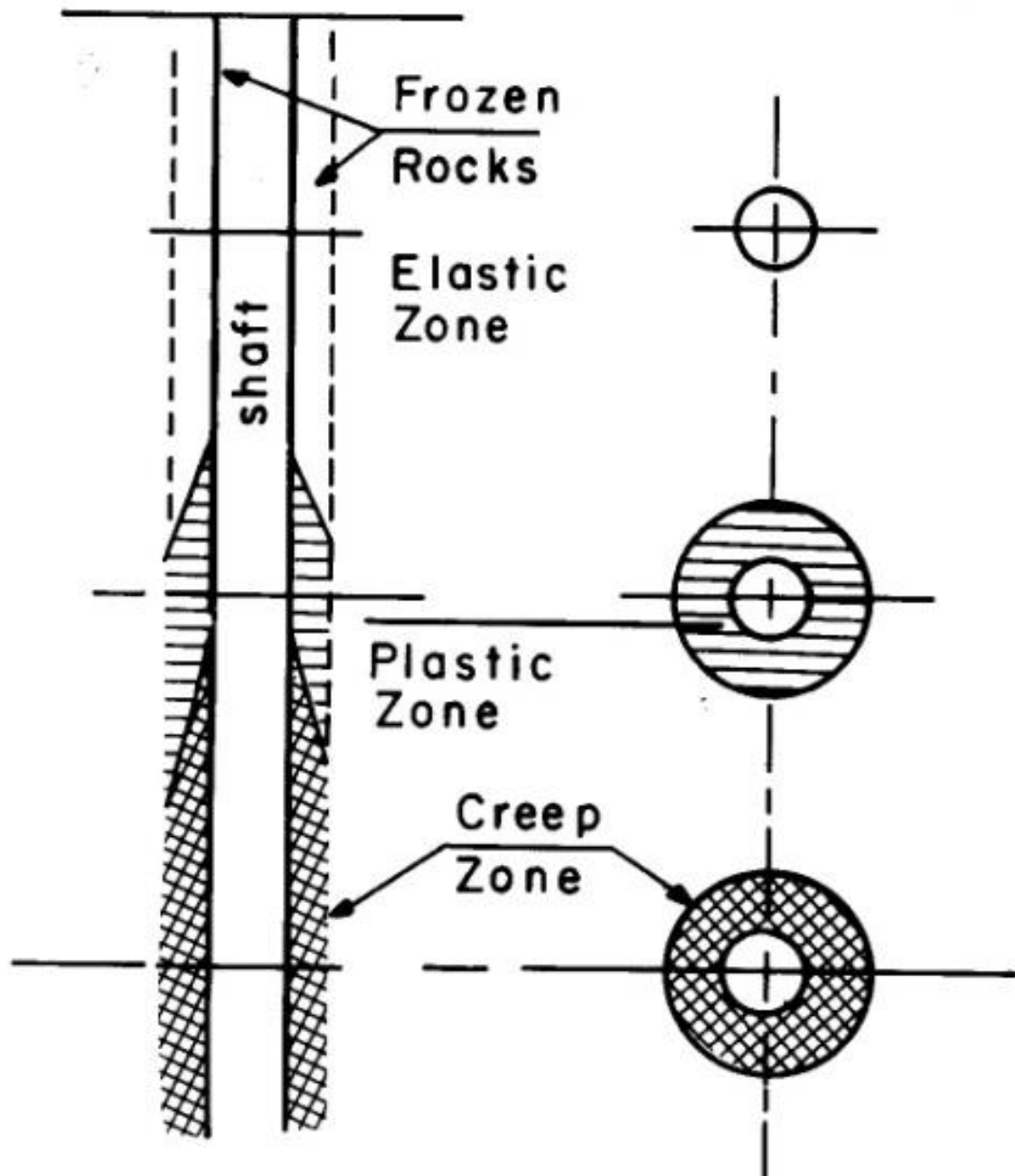
T> thickness of frozen rock cylinder

T_{ad}> adequate thickness of frozen rock cylinder to stand exposure time & creep.

D > density t/m³

H> depth of frozen rock m

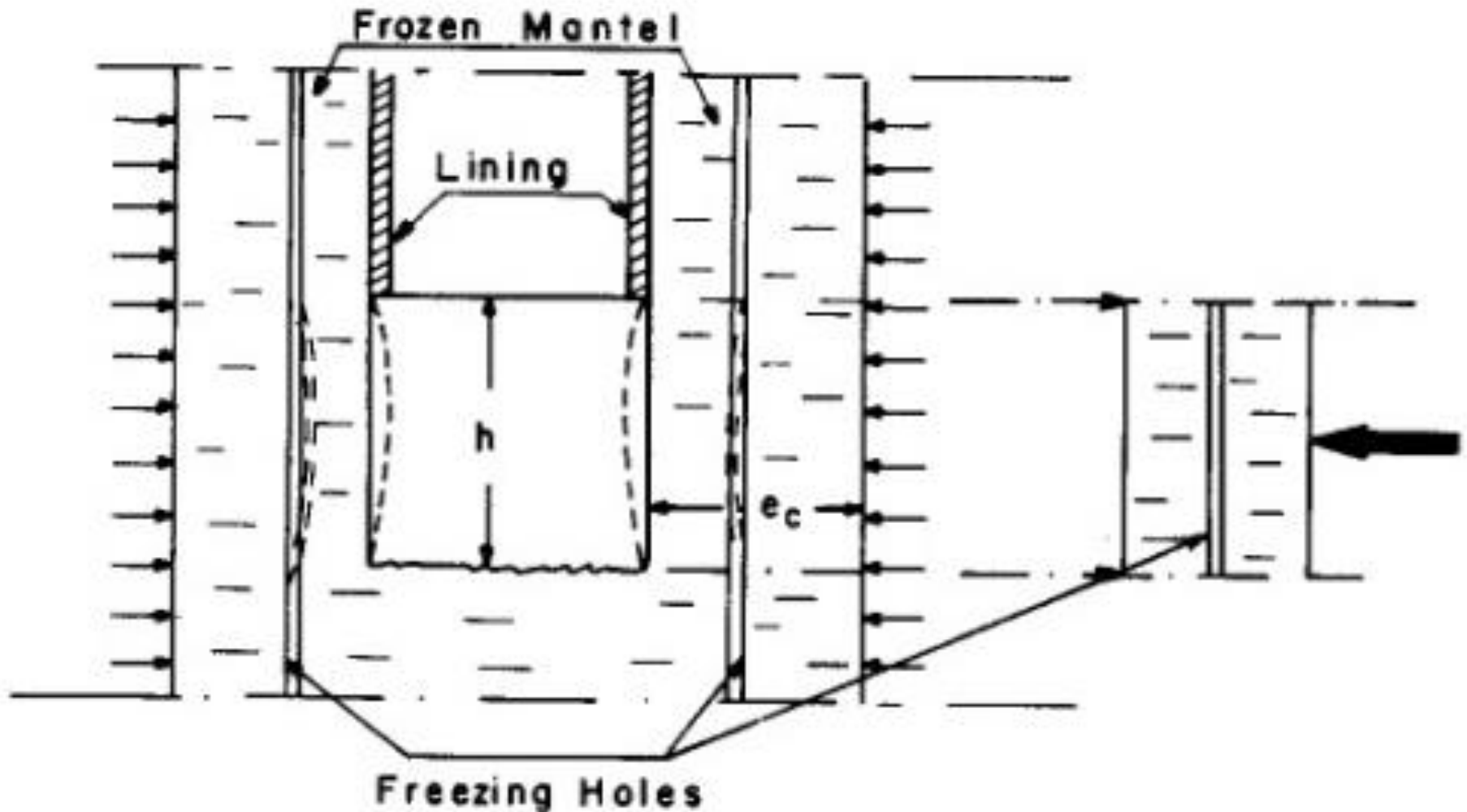
Stress in Frozen Rock



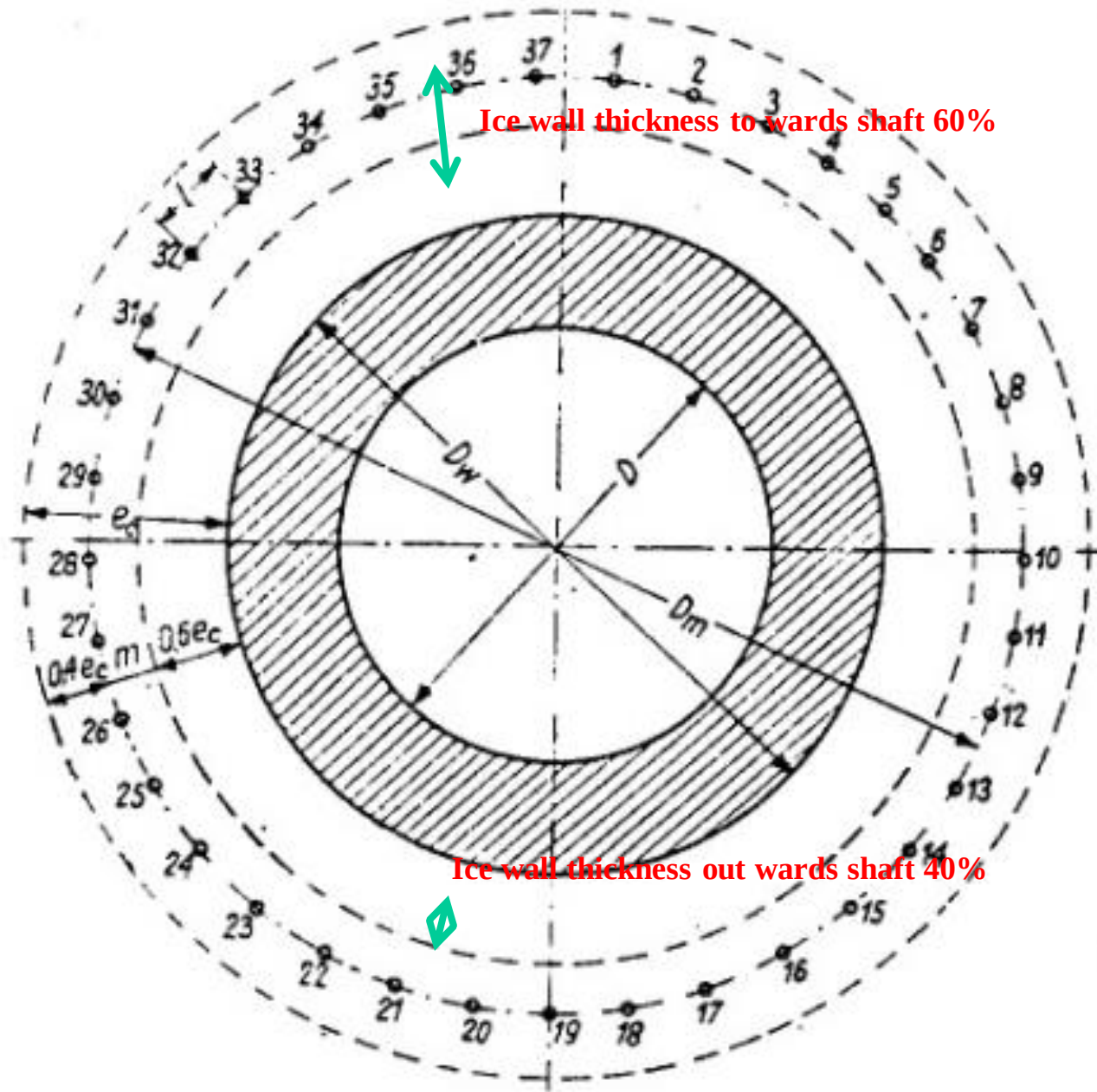
State of stress in cylinder of frozen rock with growing depth.

Shaft By Freezing I

Creep (bending) on frozen cylinder wall at great depth.



Arrangement of Freezing Hole



Arrangement of freezing holes around shaft.

D – diameter of shaft.

D_w – diameter of shaft with lining

D_m – diameter of center of freezing holes.

Freezing Hole Circle I

Centre of holes are placed on a circle with diameter calculated as below

$$D_m = D_i + 2(0.6T + m)$$

D_m > centre of holes placed on circle diameter.

D_i > shaft excavated diameter

T > thickness of frozen cylinder

m > correction for vertical deviation of freezing hole

Freezing Hole Circle I

- *Growth of frost cylinder is more towards center of shaft (0.6) than towards boundary, where heat loss is more (0.4).
- *In practice, diameter D_m is at least 2m more than shaft excavation diameter with freezing depth 80m.
- *For a depth of 150m it is 2.5m.
- *For depth below 450m, it is 3m.
- *It is possible to drill freezing holes at 500m depth with a cylinder with 1m radius.

Distance Between Freezing Hole I

- *Spacing of freezing hole varies from 0.9 to 1.1m.
- *For deep freezing with turbulent flow of brine, spacing can be increased to 1.25m.
- *Hole spacing should be arranged such that chord of over lapping circles should be equal to thickness of cylinder of frozen rock.
- *Diameter of frozen column around freezing holes at temperature (-15) degree is assumed.
- *Sandy rock diameter is 3m.
- *Clay & silt rocks is 2.5m.
- *Rock with water flow & large co-efficient of filtration – 2m.

Distance Between Freezing Hole II

*Number of freezing holes are calculated

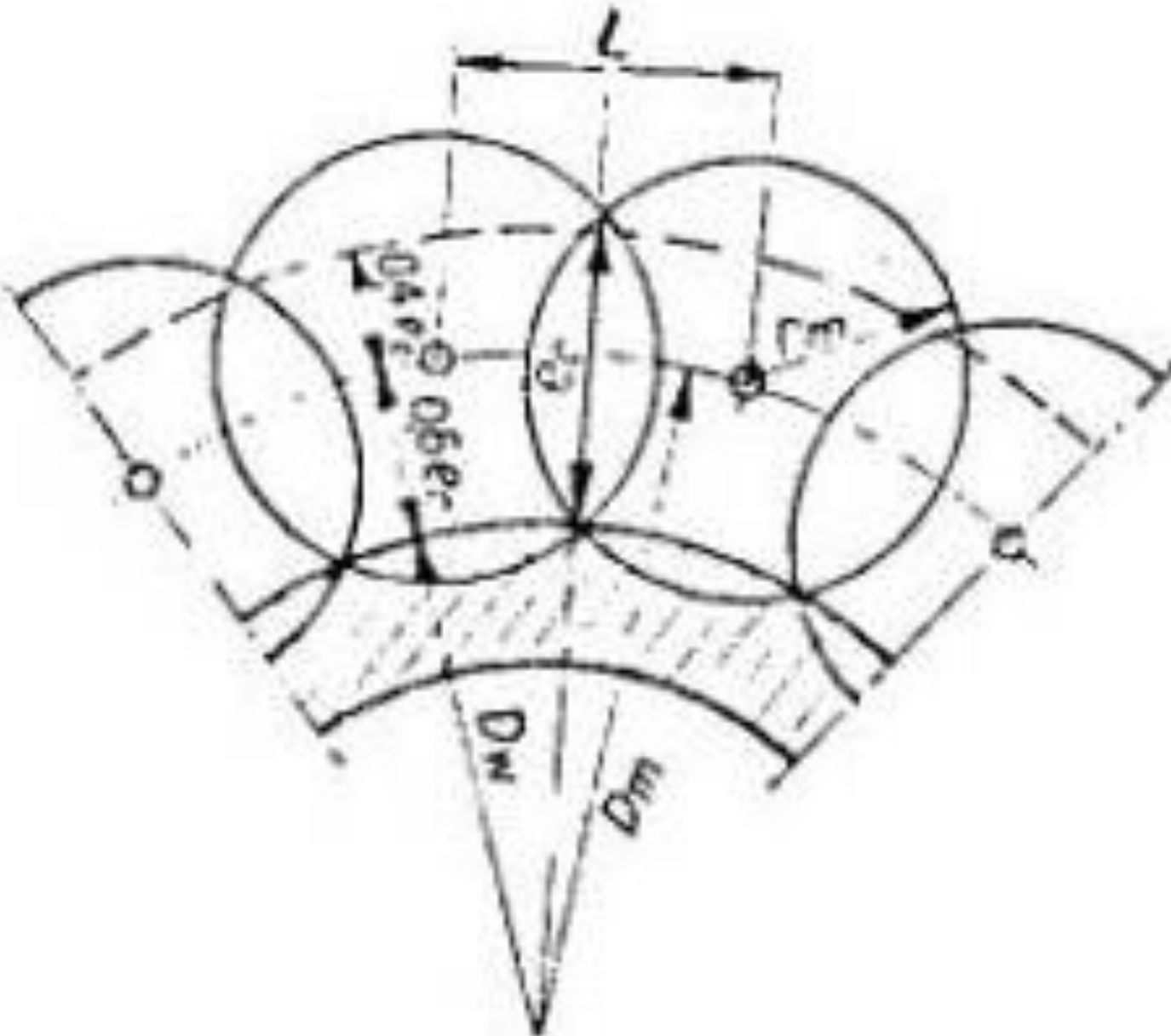
$$n = \pi D_m / l$$

n > number of freezing holes

D_m > center of holes placed on circle diameter

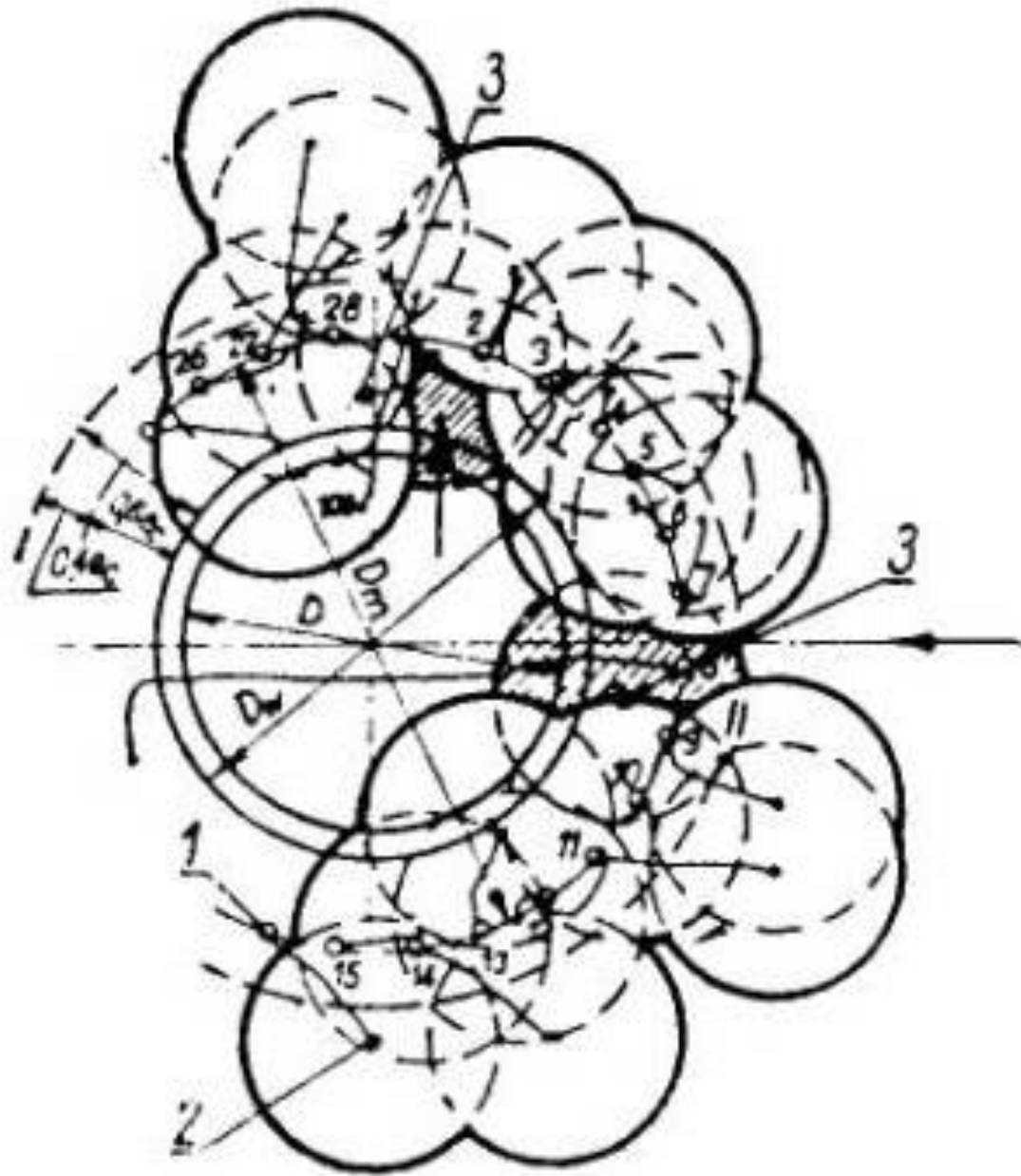
l > distance between two freezing holes

Distance Between Freezing Hole



L is distance between two freezing holes. To make freezing wall thick, holes are to be brought closer.

Map of Frozen Rock Cylinder



Map of frozen rock cylinder with freezing hole & there deviation on control horizon.

1- inlet of freezing holes.

2- bottoms of freezing holes.

3-additional holes

Freezing Hole Depth I

- *When freezing depth does not exceed 55m, bottom of hole should reach 2 to 5m below water bearing strata.
- *In inclined strata or when total depth is more, holes are drilled 5 to 10m below it.
- *Verticality of freezing hole is measured every 50m & results are plotted on a plan of whole cylinder of frozen rock.
- *If deviation of holes reduces thickness of frozen cylinder, then drilling of additional hole are planned.
- *Emphasis should be put on accuracy of drilling freezing hole.

Freezing Hole Depth II

*This shall reduce requirement of number of additional holes.

Control Holes I

- *To provide information about freezing process, additional control holes are drilled along a radial line drawn in between two freezing holes.
- *Control holes are drilled 0.5, 0.9 & 3m from freezing hole circle to wards out side & 0.4m, 1.3m towards shaft centre.
- *Holes are filled with brine & used for temperature measurement at different levels.
- *Holes are cased in steel pipe with welded bottom.
- *Water level observation hole is drilled inside frozen cylinder, usually in center of shaft.

Control Holes II

- *It has a casing with out bottom.
- *At levels, corresponding with water table, it has perforation with filter screen.
- *Observation of changes in water table are taken in hole.
- *This hole also plays rock in stress relief, allowing water entrapped with frozen cylinder to flow up in to hole.
- *Otherwise, volumetric changes during freezing process will increase hydrostatic pressure.

Drilling Of Freezing Hole I

- *Most important problem during freezing to assure verticality of holes.
- *Modern rotary drilling rigs, when operated by competent operator, can maintain verticality with in acceptable tolerance.
- *Freezing holes are drilled with mud fluid to specific gravity 1.15 to 1.25.
- *To shorten time requirement for drilling of many (some time up to 40) holes, 2 or 3 rigs are used at a time.

Freezing Hole Equipment I

- *Every hole is equipped with two column of tubing.
- *Outer Tube – freezing pipe it self has diameter of 102 to 152mm (4 to 6 inch) & is made of steel with inside connecter so it has smooth outside.
- *Outer pipe has welded bottom.
- *Inner tube – has smaller diameter 32 to 51mm (1.75 to 2 inch), is made of steel pile.
- *Polyethylene pipe is also used as inner pipe.
- *Polyethylene pipe has inside diameter 75mm & wall thickness 43mm.

Freezing Hole Equipment II

- *Polyethylene is cheaper & easier to install by unwinding from reel.
- *Polyethylene has lower thermal conductivity.
- *Specific gravity of Polyethylene pipe is very close to specific gravity of brine solution, resulting less suspended weight.
- *Freezing brine flow at higher velocity, down in inner pipe.
- *Brine solution moves slowly through annular space, absorbing heat from surrounding rock.

Freezing Hole Equipment III

- *There is two distinct flow regimens, laminar & turbulent.
- *For greater depth & more difficult freezing condition associated with presence of clay rocks & organic matter (lignite), turbulent flow is recommended.
- *It assures more uniform heat flow from rock to freezing hole, making it less dependent on depth.
- *Laminar flow is adequate for moderate depth, exhibits a distinct difference in velocity of brine flowing down 0.6 to 1.5m/s (2 to 4.5 ft/s) & upward 0.08 to 0.2m/s (0.25 to 0.67 ft/s).

Freezing Hole Equipment IV

- *Brine circulation initially cools rock & latter freezes rock creating rigid column of 2 to 3m(6 to 9ft) in diameter.
- *There is large difference in thermal conductivity of rocks, resulting in difference in freezing time, varying from 2 to 8 months.
- *For greater freezing depth, larger diameter freezing pipes are required.
- *Ratio of diameter between outer & inner has to be chosen for particular depth & existing heat transfer rate.

Freezing Hole Equipment V

- *This keeps cost of energy (freezing compressor, pump) to a minimum & also ensures an installation capacity of freezing rock in scheduled time interval.
- *Complete tightness of joints among individual segment of freezing pipes is necessary.
- *Any brine leak will thaw rocks & in extreme cases, can cause catastrophic inflow of outside water & rock into shaft.
- *It is impossible to freeze rock contaminated with brine with out substantial lowering temperature

Freezing Hole Equipment VI

- *Possibility of lowering temperature for a particular frizzing station is limited.
- *During installation of freezing pipes, every column of 50m (150ft) or longer is to be hydrostatically tested.
- *When whole column is lowered to its final position, casing pipe are removed.
- *Space between wall of hole & freezing pipe is then filled with dense drilling mud.

Freezing Hole Equipment

CONSTRUCTION OF

Freezing brine circulation in freezing holes.

1- freezing cellar lining

2- distribution ring

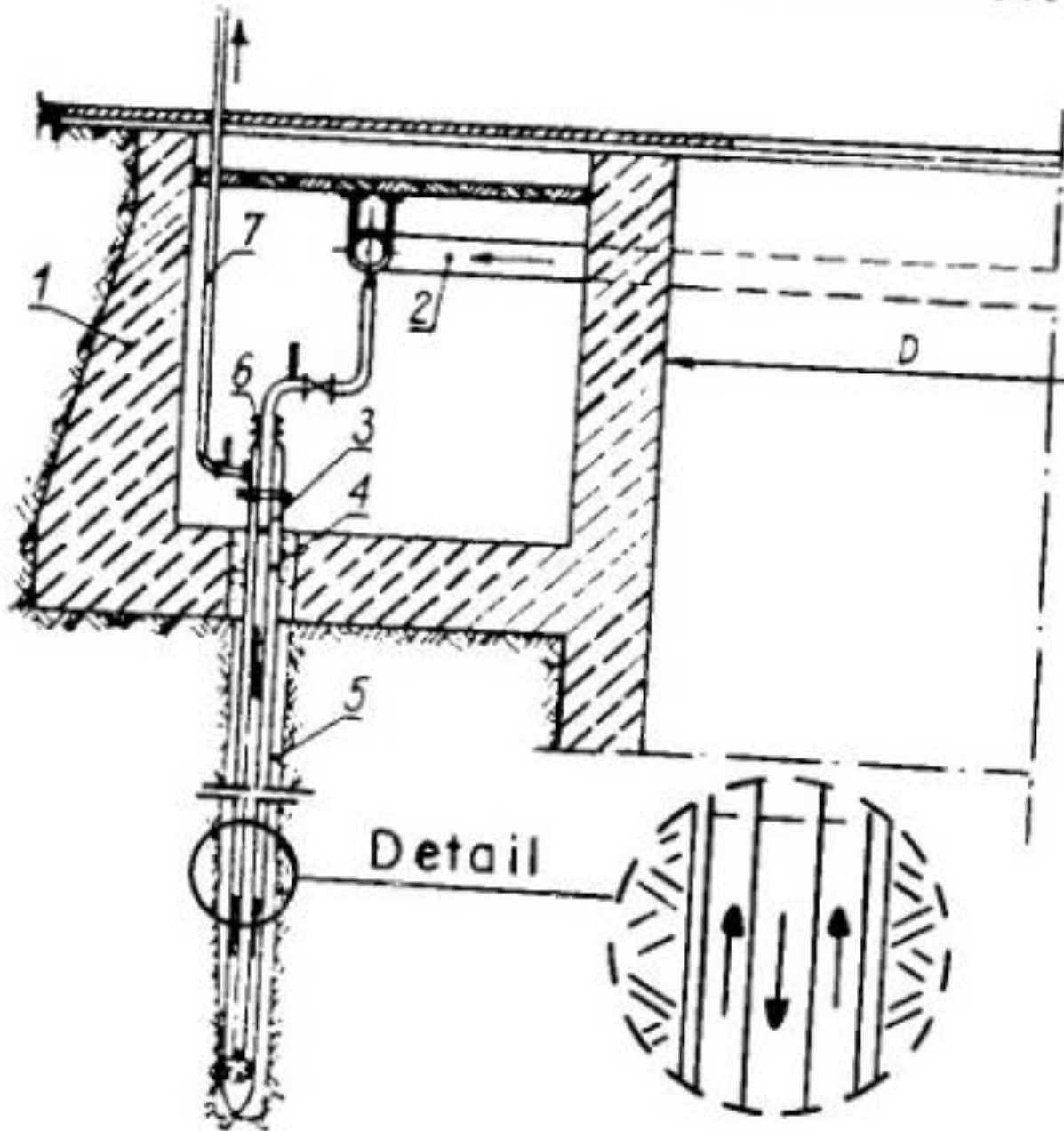
3-entrance tubing

4-freezing pipe

5-freezing hole

6-tee

7- out flow pipe to inflow brine tank



Sectional Freezing I

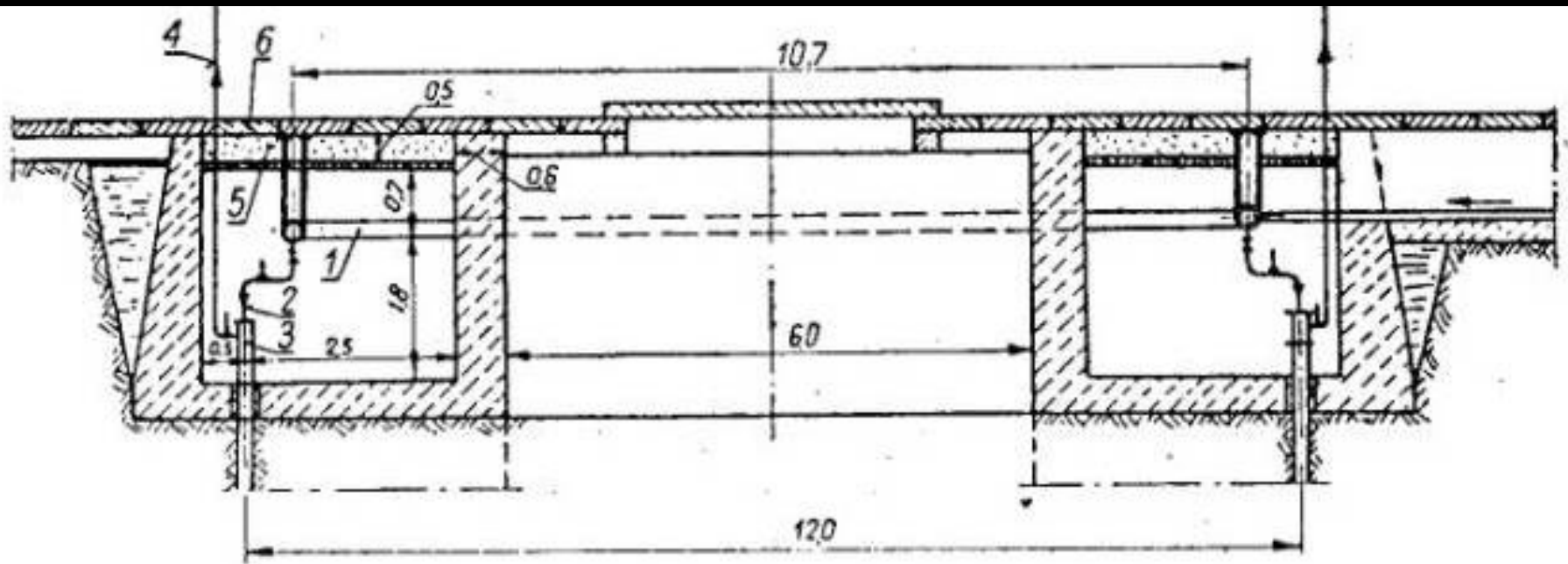
- *Some time water bearing strata under lies strata of competent rocks.
- *In this case, freezing is required only at particular depth.
- *A brine circuit is designed to enable control of return flow from particular point.
- *Alarm gauge in tank gives warning signal, when brine level goes down, indicating a break of freezing pipe & escape of brine into rock mass.
- *Such hole has to be immediately shut off (valve inflow line).

Freezing Cellar I

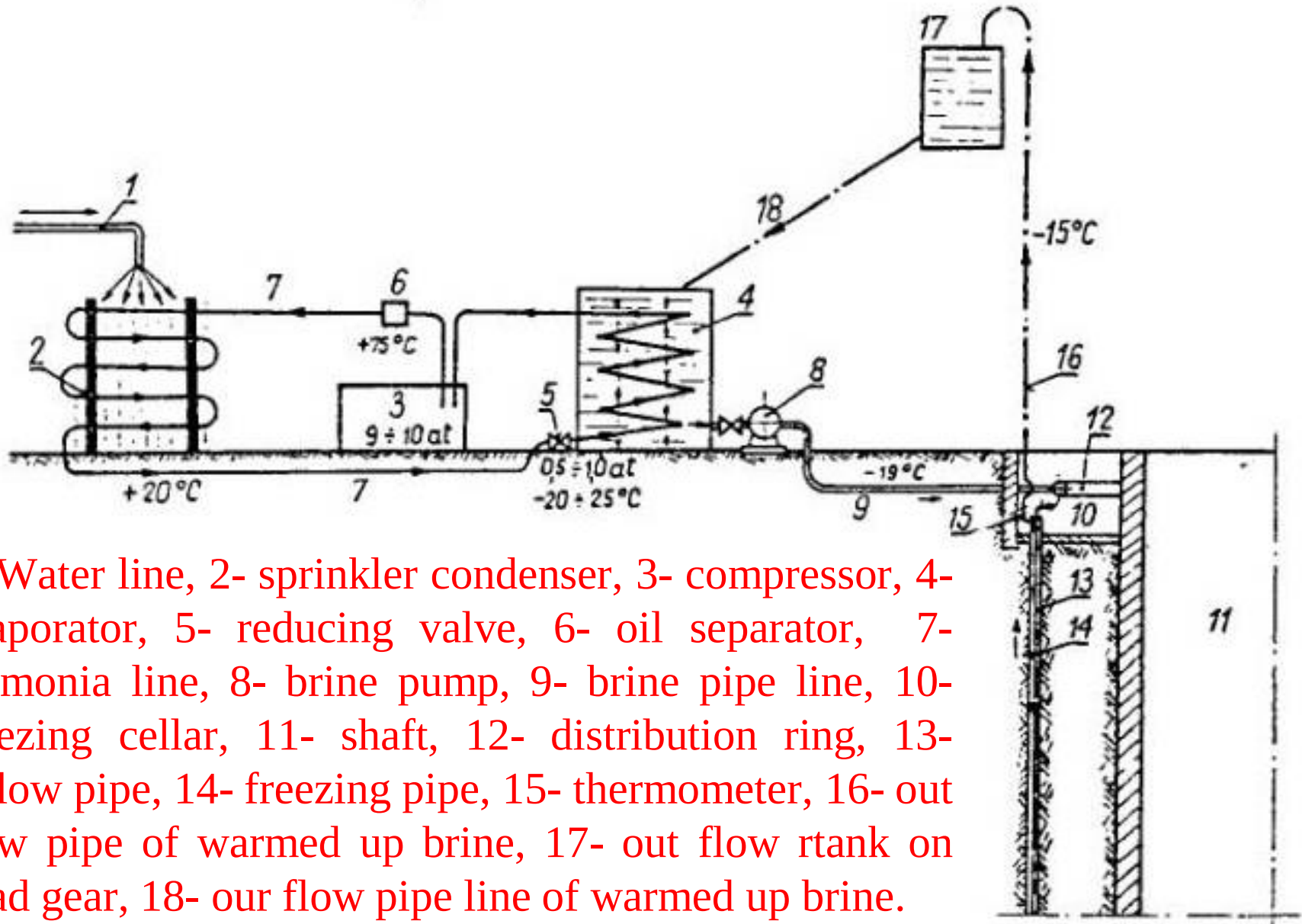
- *Cellar of freezing hole with pipes connecting them with distribution ring, shut off valve & thermostat are installed all together in freezing cellar.
- *Freezing cellar is temporary to startup construction time & will be demolished when permanent shaft head frame is installed.
- *Permanent cellar is part of permanent shaft head frame requiring larger construction time.

Freezing Cellar

1- distribution ring, 2- inflow pipes (supply cold brine), 3- freezing pipe, 4- pipe with return warm brine, 5- beam floor with thermal insulation, 6- surface beam floor



Active Freezing Time I



1- Water line, 2- sprinkler condenser, 3- compressor, 4- evaporator, 5- reducing valve, 6- oil separator, 7- ammonia line, 8- brine pump, 9- brine pipe line, 10- freezing cellar, 11- shaft, 12- distribution ring, 13- inflow pipe, 14- freezing pipe, 15- thermometer, 16- out flow pipe of warmed up brine, 17- out flow rtank on head gear, 18- our flow pipe line of warmed up brine.

Freezing Rock Mass I

- *Ammonia NH_3 is used as refrigeration medium for cooling down brine solution.
- *Ammonia assures a freezing temperature of $-20\text{ }^\circ\text{C}$ with one stage & $-40\text{ }^\circ\text{C}$ two stage of compression.
- *Ammonia is most used media.
- *Carbon dioxide CO_2 provides a freezing temperature of $-50\text{ }^\circ\text{C}$.
- *In one compression freezing station compressor raise pressure of ammonia gas to 1Mpa with simultaneous increase of temperature to $75\text{ }^\circ\text{C}$.

Freezing Rock Mass II

- *Compressed gas goes through oil separator.
- *Compressed ammonia gas go to sprinkler condenser.
- *At condenser, ammonia gas is cooled down with water of 12 C to temperature around 20 C.
- *At this temperature gaseous ammonia changed into a liquid.
- *Liquid ammonia passes through expansion valve to a pressure reducer. to
- *Pressure is reduced from 1Mpa to 50 to 140 kpa.
- *Due to rapid expansion, temperature is reduced to -20 C.

Freezing Rock Mass III

- *Vapor of ammonia travels through coil pipe with evaporator, filled with brine.
- *In this process, brine cools down.
- *Then ammonia return to intake side of compressor & cycle starts again.

Brine Circuit I

- *A water solution of calcium chloride, CaCl_2 , is used almost exclusively for temperature not lower than -50 C.
- *Most often used aqueous solution is of CaCl_2 with specific gravity of 1.25 to 1.26 & freezing temperature -34 to -38 C.
- *Brine circuit in contact with rock get heated up to 3 C.
- *Hot brine returns to a collector ring or over flow tank on shaft then to main pipe line.
- *Brine ultimately gets cool down in heat exchanges before going down hole in underground.

Freezing Period I

- *There are two freezing period, active & passive (maintaining period).
- *In active period, a frozen mantel of design thickness is created around future shaft.
- *To achieve designed capacity of freezing station 2 or 3 units are installed to provide required heat efficiency.
- *During active period, all unites work at full capacity.
- *Passive period starts after frozen mantel is created & last through sinking, lining & assembling if shaft installation.

Freezing Period II

*During passive period, usually only one unit works while other stands by as reserve.

Heat balance I

- *Volume of frozen rock has to be found.
- *Volume is calculated as sum of cylinder in particular strata of different physical parameters.
- *Volume of each strata is sum of volume of rock material (solid) & water volume remaining in rock pores
- *Water content of rock depends on rock property, fracture & pore volume.
- *Required data is obtained from rock test.

Heat balance II

Heat balance formula

$$Q_0 = Q_1 + Q_2 + Q_3 + Q_4$$

Q_0 > heat to be extracted to cool down rock mass from initial temperature T_i to mean temperature.

Q_1 > heat to be extracted from rock mass to lower initial temperature of solid to mean freezing temperature.

Q_2 > heat to be extracted to cool down water in pores from initial temperature to 0 C.

Q_3 > heat to be extracted to change water at temperature 0 C to ice at constant temperature.

Heat balance III

Heat balance formula Cont>

Q4> heat to be extracted to cool down ice at initial temperature 0C to final freezing temperature.

Unit of heat is kcal.

*Mean initial rock & water temperature is approximately is 10 to 12 C (cold countries).

*Above heat component are calculated by applying calorimetric formula, that takes following in to accounts

*Dimension of particular rock cylinder.

*Corresponding content of solid.

Heat balance IV

- *Heat capacity of rock, water.
- *Latent heat of change water in to ice.
- *Heat & density of ice.
- *Temperature difference achieved by freezing.
- *Required neat capacity of freezing station can be calculated.
- *Heat loss in rock mass & installation are to be calculated.
- *Size of plant depends on amount of heat to be extracted during active freezing period.

Heat Loss I

- *Heat loss takes place due to steady flow of heat from out side frozen mantel.
- *Loss in freezing installation because of imperfect insulation.
- *Gross capacity of freezing plant station is equal to its neat capacity minus heat loss in rock mass, plus heat loss in equipment.

Active Freezing Time I

- *Time of active freezing of a rock mass depends on rate frost penetration radiating out ward from hole to reach desired / calculated radius (R).
- *Radius (R) should assures sufficient thickness of frozen mental, measured between two holes.
- *Frost penetration velocity for sand is about 25mm/day (1in/day), for clay it is about 10-15mm/day (0.2 to 0.25in /day).
- *Freezing velocity in practical term for given set of rock mass parameter & hydrologic condition depends mainly on neat capacity of freezing station.

Active Freezing Time II

- *As per field experience, time of active frizzing is about 2 to 4 months for freezing depth 250m.
- *For 500 m freezing depth active freezing time is 9-10 months.
- *Active freezing time for a frozen mantel will depends on
 - *Brine temperature
 - *Spacing of freezing holes.
 - *Initial temperature.
 - *Co-efficient of thermal conductivity.

Passaive Freezing Time I

- *In passive freezing period, already frozen mantel of rock mass should be maintained at designed freezing temperature during entire period of sinking & lining of shaft.
- *Required capacity of freezing station during this period is smaller, station should balance heat loss.
- *In general, during passive period, one half to one third of initial capacity is sufficient to keep rock mass frozen.
- *Computer numerical modeling method are used for heat transfer calculation for design of shaft sunk with freezing method.

Excavation In Frozen Ground I

- *Method of shaft excavation is similar to that of normal shaft excavation.
- *Specific care has to be taken while using explosive in shaft sinking.
- *Care should be taken not to damage freezing holes.
- *When core of shaft is not frozen a cactus grab or other mechanical mucker can dig shaft bottom.
- *When rock is partly frozen and too hard to be excavated, hand held drill hammer with chisel pick are used.

Excavation In Frozen Ground II

- *Working condition at bottom of shaft is not healthy for manual work.
- *In hard frozen rock, drilling blasting method is used.
- *In drilling blasting, minimum distance between freezing hole & blast hole should be 1.5m.
- *Depth of blast hole should not exceed 1 to 1.7m (3 to 6ft).
- *In shaft of larger diameter, hole density can be 0.5 to 1.5hole/m²
- *Estimated explosive consumption in alluvial overburden 0.25 to 0.5kg/m³ & 1 to 1.4 in sandstone, sandy shale & lime stone.

Excavation In Frozen Ground III

- *In latest excavation system, specially designed miner (similar to road header, or coal cutter) is being used.
- *This system avoids human exposure at frozen shaft bottom rock.

Reference I

*SME Mining Engineering Hand Book, 2nd Edition,
Volume 2, Page 1610 to 1621